

PROJECT DESIGN PHASE-II : DATA FLOW DIAGRAM & USER STORIES

SZ METRO TICKET GENERATING SYSTEM IN SERVICENOW

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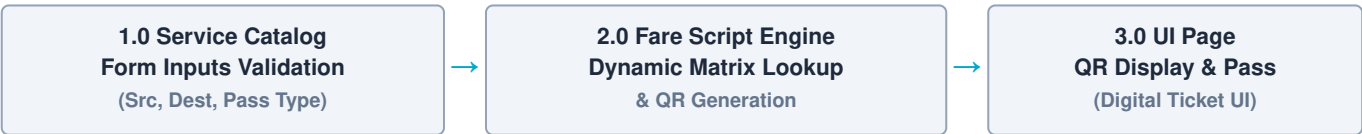
1. DATA FLOW DIAGRAMS (DFD LEVEL-1)

A Data Flow Diagram (DFD) provides a traditional visual representation of information flows within the system. A neat and clear DFD depicts system requirements graphically by illustrating how data enters and leaves the system, what processes transform the information, and where data is persistently stored.

System Data Flow & ServiceNow Execution Architecture

- User Configuration & Form Input:** Passengers interact with ServiceNow Service Catalog dynamic variables (Source, Destination, Passenger Type, Ticket Count).
 - Automated Calculation Engine:** Business Rules and Client Scripts trigger dynamic fare band evaluation based on station matrix tables.
- QR Code Engine:** System generates a unique encrypted QR payload encoding ticket ID, travel route, timestamp, and validity.
 - Persistence & Visualization:** Data is logged into custom ServiceNow tables (x_sz_metro_ticket), rendering a digital ticket UI with flow logs mapped for administrative tracking.

DFD LEVEL-1 EXECUTION PIPELINE



Persistent Data Stores: [D1: Station & Fare Tables] • [D2: Ticket Logs & Users Table (`x\_sz\_metro\_ticket`)]

2. IMPLEMENTATION METHODOLOGY (WHAT → WHY → HOW → OUTCOME)

WORKFLOW EXECUTION PATH	TECHNICAL IMPLEMENTATION & EXECUTION STRATEGY
What	Dynamic Catalog Form & Dynamic Fare Band logic configured in ServiceNow.
Why	Eliminates manual fare entry, minimizes human error, and ensures accurate, automated real-time pricing.
How	Configured ServiceNow Catalog Client Scripts & UI Policies integrated with backend Script Includes.
Outcome & Verification	Outcome: Real-time QR generation with annotated visual feedback for passenger mobile display. Data Storing: Structured table schema mapping ticket attributes for auditability. Future Scope: Integration with SMS/Email gateways & Executive Analytics Dashboards.

3. USER STORIES & ACCEPTANCE CRITERIA

The following table outlines the comprehensive user stories, acceptance criteria, release milestones, and priority levels for the product:

USER TYPE	FUNCTIONAL REQUIREMENT (EPIC)	STORY #	USER STORY / TASK	ACCEPTANCE CRITERIA	PRIORITY	RELEASE
Passenger (Mobile / End User)	Ticket Booking	USN-1	As a user, I can select Source, Destination, Passenger Type, and Ticket Count in the Service Catalog form.	Form dynamically displays valid station pairs and fields adjust on selection.	High	Sprint-1
Passenger (End User)	Fare Calculation	USN-2	As a user, I want dynamic automated fare calculation based on route distance and passenger type.	Fare is instantly computed using dynamic fare bands and updated on UI.	High	Sprint-1
Passenger (End User)	Digital Ticketing	USN-3	As a user, I can receive a digital ticket with an embedded scannable QR code upon booking confirmation.	Scannable QR code generates on UI Page displaying valid ticket information.	High	Sprint-2
Administrator (IT Admin / Dev)	Backend Configuration	USN-4	As an Admin, I can configure dynamic fare logic scripts, station tables, and catalog setup in ServiceNow.	All setup steps executed cleanly with full flow log execution tracking.	High	Sprint-1
Administrator (IT Admin)	Data Persistence	USN-5	As an Admin, I can store passenger ticket transactions securely in a dedicated custom table schema.	User data, ticket ID, fare, and timestamp stored in custom table schema.	Medium	Sprint-2
Station Manager (Operations)	Validation & Audit	USN-6	As a Station Manager, I can view active ticket logs and verify passenger QR codes at gates.	Manager dashboard accurately reflects active ticket statuses and scan logs.	Medium	Sprint-2
Operations Lead	Travel Analytics	USN-7	As an Operations Lead, I can view visual dashboards showing peak travel routes and ticket volumes.	Real-time reports display traffic trends using realistic sample data.	Low	Sprint-3
Passenger / System	Multi-channel Ticketing	USN-8	As a user, I can receive my QR ticket via SMS, Email, and payment gateway integration (Future Scope).	Notification engine sends SMS/Email with ticket QR payload.	Low	Sprint-3